

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.2 LINEAR MOTION: FINAL EXPLANATION – THE MKOKOTENI MYSTERY SOLVED

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation. Think back to the runaway mkokoteni loaded with sacks of maize on the Limuru–Nairobi escarpment road. Answer every question in the space under it, working on your own. For calculations, write GIVEN, FIND, the equation you choose, your working and your answer with units. Use $g = 10 \text{ m/s}^2$ for free fall. Use the scientific words you learned (distance, displacement, speed, velocity, acceleration). Write in full sentences when you are asked to explain.

Part 1: Why does the mkokoteni keep getting faster?

The mkokoteni was released from rest at the top of the Limuru escarpment. Nobody pushed it and it has no engine, yet witnesses said it became dangerously fast within seconds.

(a) Some people in your Lesson 1 discussion said 'something must be pushing it'. Use your knowledge of gravity to explain what is really happening to the cart's motion.

(b) What do we call the continuous change in velocity of the cart? Explain how it is different from speed.

(c) Explain why the cart's speed keeps increasing the whole time it rolls downhill, rather than increasing once and then staying the same.

Write your answer here:

Part 2: Predicting speed, distance and time with the equations

Your class found that the mkokoteni's constant acceleration down the escarpment is about 3 m/s^2 . It starts from rest ($u = 0$).

- (a) Which equations of linear motion would you choose to find its speed and its distance after 4 s? Write the GIVEN and FIND quantities first, then calculate both.
- (b) Witnesses said the cart was moving at about 40 m/s when it crashed. How long had it been rolling at that moment? Show your working.
- (c) Use $v^2 = u^2 + 2as$ to find how far down the road the cart had travelled when it reached 40 m/s. Give your answer to the nearest metre.
- (d) Explain, using your answers, why we can say we CAN predict how fast and how far the cart travels before it crashes.

Write your answer here:

Part 3: Speed along the escarpment and a model of the cart

Use $a = 3 \text{ m/s}^2$ and $u = 0$ for the mkokoteni. Complete the table by calculating the speed v at each distance marker from the start, using the correct equation.

Distance from start, s (m)	Speed, v (m/s)
0	
50	
100	
200	

(a) Fill in the speed column (one decimal place) and show one full calculation.

(b) From 50 m to 100 m the distance doubles. Does the speed double as well? Use your table to answer and explain what this tells you about the relationship between speed and distance.

(c) You are drawing the class wall model of the mkokoteni descending the escarpment. Describe what the velocity arrows at these four markers should look like, and name two more labels or symbols you would add to the model to show how a crash could be prevented.

Write your answer here:

Part 4: Free fall – the mango and the cart

Your class dropped a rubber bung, a crumpled paper ball and a small stone from 2 m and observed what happened. Now think about a ripe mango falling from a tree near Kisumu and the mkokoteni on the escarpment. Use $g = 10 \text{ m/s}^2$.

- (a) Describe what you saw in the drop experiment when two objects of different mass were dropped together. Explain why a heavy object does not fall faster than a light one.
- (b) A mango is released from rest and falls freely for 2 s. Calculate its speed after 2 s and the distance it falls.
- (c) The mkokoteni's acceleration (about 3 m/s^2) is smaller than g . Explain how free fall is related to the mkokoteni's motion, and say why the mkokoteni's acceleration is not equal to g .

Write your answer here:

Part 5: Road safety and the Pitch to the County Engineer

In Lesson 6 you studied a matatu braking on Mombasa Road. Suppose the matatu brakes with a constant deceleration of 5 m/s^2 until it stops ($v = 0$). Use $v^2 = u^2 + 2as$ with $a = -5 \text{ m/s}^2$. Convert km/h to m/s by dividing by 3.6.

- (a) Calculate the braking distance for a matatu travelling at 30 km/h and at 90 km/h (give your answers to one decimal place).
- (b) Compare the two distances. What happens to braking distance when speed is tripled, and what does this tell you about the relationship between speed and braking distance?
- (c) Write a short pitch (4–6 sentences) to the Limuru County Engineer. Explain why the mkokoteni kept accelerating, what mathematics could have predicted the crash, and one evidence-based safety measure you recommend for the escarpment road. In your answer, answer the KICD questions: how is free fall important in real life, and how can its dangers be reduced?

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific explanation of the phenomenon	Clearly explains that gravity is a constant unbalanced force down the slope that produces constant acceleration, rejects the 'something is pushing it' idea, and links acceleration to velocity changing every second.	Identifies gravity as the cause of the increasing speed and names acceleration, but the explanation of why it is continuous or constant is incomplete.	Gives vague or incorrect reasons (for example, a push or the cart's mass alone) or confuses speed with acceleration.
Use of equations of linear motion	Lists GIVEN and FIND, selects the correct equation every time, rearranges correctly and gets correct answers with units (12 m/s, 24 m, 13.3 s, 267 m).	Selects mostly correct equations and gets most answers right, with small slips in rearranging, rounding or units.	Chooses incorrect equations, omits working or units, or cannot substitute values correctly.
Interpreting data and relationships	Completes the table accurately and explains the non-linear relationships (speed against distance, braking distance against speed) using the equations and the 1.4 times and nine times comparisons.	Completes the table and recognises that the relationships are not straight-line, but gives limited supporting numerical evidence.	Table has several errors or the student assumes that doubling distance or tripling speed simply doubles or triples the outcome.
Free fall and the mass question	Uses observations from the drop experiment, explains with $a = F/m = g$ why mass does not matter, and links free	States that all objects fall with the same acceleration and calculates free-fall values correctly, but the explanation of	Says heavier objects fall faster or gives no reason for equal acceleration.

	fall to the mkokoteni as a case of the same gravitational acceleration.	why mass does not matter is brief.	
Real-life application and communication	The pitch is clear and well organised, uses Kenyan road-safety evidence and calculated values, gives a sensible mitigation measure, and answers both KICD questions using correct scientific vocabulary.	The pitch covers the explanation and one safety measure, with some use of numbers and vocabulary.	The pitch is unclear or incomplete, gives no safety measure, or does not link the mathematics to the real situation.

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